



Conversion of an obligate autotrophic bacteria to heterotrophic growth: expression of a heterogeneous phosphofructokinase gene in the chemolithotroph *Acidithiobacillus thiooxidans*

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Abstract

A plasmid pSDK-1 containing the *Escherichia coli* phosphofructokinase-1 gene (*pfkA*) was constructed, and transferred into *Acidithiobacillus thiooxidans* Tt-7 by conjugation. The *pfkA* gene from *E. coli* could be expressed in this obligately autotrophic bacterium but the enzyme activity (18 U g^{-1}) was lower than that in *E. coli* (K12: 86 U g^{-1} ; DF1010 carrying plasmid pSDK-1: 97 U g^{-1}). In the presence of glucose, the Tt-7 transconjugant consumed glucose leading to a better growth yield.

Introduction

Acidithiobacillus thiooxidans is an acidophilic, chemolithotroph bacterium which derives its energy by oxidizing elemental sulfur or reduced sulfur compounds and obtains its carbon by fixing CO_2 . It has been used industrially in metal leaching and the microbial desulfurization of coal in combination with *A. ferrooxidans*. However, the slow growth rate of this organism has limited its further use. Recent studies showed that some genes, such as isocitrate dehydrogenase (ICDH) gene (*icd*), from *A. thiooxidans* can function in *E. coli* (Inoue *et al.* 2002). Some plasmids have been transferred successfully from *E. coli* to *A. thiooxidans*, and genes originating from heterotrophic bacteria could be expressed as well (Jin *et al.* 1992, 1993, Shao *et al.* 1997). It is therefore possible to construct new *A. thiooxidans* strains with better growth rates and increased leaching efficiencies.

Phosphofructokinase (EC 2.7.1.11) is a key enzyme in the control of glycolysis. Phosphofructokinase, as well as some other enzymes such as 6-phosphogluconate dehydrase and α -ketoglutarate dehydrogenase, are missing in *A. thiooxidans*. This microorganism therefore cannot grow on organic nu-

trients (Matin 1978). In this paper, a plasmid, pSDK-1 containing *E. coli* *pfkA* gene, was constructed and transferred into *A. thiooxidans*. The expression of the *pfkA* gene and the effect of glucose on the growth of this new organism were studied. This work would provide a new insight into how to improve the growth rate and leaching performance of the autotrophic bacteria.

Materials and methods

Bacterial strains, plasmids and growth conditions

The bacterial strains and plasmids used in this study are listed in Table 1. *E. coli* DF1010 lacking phosphofructokinase activity cannot grow on mannitol but can grow on glycerol. *E. coli* DF1010 with a plasmid carrying an active phosphofructokinase gene can be selected by genetic complementation on mannitol-selective plates. Strains of *A. thiooxidans* were grown in Starkey-S⁰ inorganic medium at 30 °C as described previously (Jin *et al.* 1992). The solid medium was prepared by mixing component A (double strength basal salts and agar, pH 4.8, autoclaved) and B (sodium thiosulfate, 2%, w/v, filter-sterilized) in equal volumes before using. This medium was supple-

Table 1. Bacterial strains and plasmids.

Strain or plasmid	Phenotype or genotype	Source or reference
<i>E. coli</i> strains		
K12	Wild type	Stock Center of China,
DF1010	<i>garB10, fhuA22, ompF627(T2R), ΔpfbB201, fadL701</i> (T2R), <i>recA56, relA1, pit-10, spoT1, Δ(rhaD-pfkA)</i> 200, <i>rrnB-2, creC510</i>	Beijing <i>E. coli</i> Genetic Stock Center, Yale University
SM10	<i>thi, thr, leu, tonA, recA, lacY, supE, Mu_c⁺</i> (RP4-2Tc::Mu)	Simon et al. (1983)
<i>A. thiooxidans</i>		
Tt-7	Wild type	Jin et al. (1992)
Plasmids		
pJRD215	Km ^r Sm ^r IncQ <i>mob</i> ⁺	Davison et al. (1987)
pSDK-1	Km ^r IncQ <i>mob</i> ⁺ <i>pfkA</i> ⁺	This study

mented with 0.05% (w/v) yeast extract when used as a mating medium for *E. coli* and *A. thiooxidans*.

Construction of plasmid pSDK-1 containing *pfkA* gene

For the amplification of *pfkA* gene, sense primer (5'-GTACGGATCCTTGGCCTGACCTGAATCAAT-3', *Bam*HI) and antisense primer (5'-GCGCAAGCTTCCGACTCTCTTATGTTGTGT-3', *Hind*III) were synthesized. Total DNA of *E. coli* K12 was extracted and used as template for PCR of *pfkA* gene. PCR reactions were performed as follows: 5 min at 94 °C, 30 cycles of 1 min at 94 °C, 1 min 30 s at 50 °C, 2 min 30 s at 72 °C, and finally 10 min at 72 °C. The 1.4 kb PCR amplified fragments, which contain the promoter and coding sequence of *pfkA* gene, were purified and cloned into the pJRD215 digested with *Bam*HI and *Hind*III to construct the plasmid pSDK-1. This plasmid was then transformed into *E. coli* DF1010 and screened by complementation on mannitol-selective medium. The inserted fragment in pSDK-1 was sequenced by the dideoxy chain termination method, which was performed with Beckman CEQ2000XL DNA Analysis System.

Conjugation

Donor cells were harvested by centrifugation at the late exponential growth phase; recipient cells were harvested at the stationary phase. Sulfur precipitates were removed from the liquid culture by centrifugation at 800 × *g* for 5 min. Crosses between strains were conducted by filter mating by using a 1:1 mixture of donor and recipient cells that had been suspended

in saline (pH 4.6–4.8). After incubation at 30 °C for 48 h, the cultures were washed from the filter, diluted, plated on appropriate selective plates and nonselective plates and incubated at 30 °C for 7 d. As a control for spontaneous mutation, both parental strains were plated on the same selective plates. The transfer frequency of conjugation was calculated by the number of transconjugants divided by that of recipients.

Stability of pSDK-1 in *A. thiooxidans*

Single colonies of *A. thiooxidans* transconjugants on the selective plates were transferred into antibiotic-free Starkey-S⁰ medium. After incubating at 30 °C for 7 d, 1% (v/v) of the fully grown cultures was transferred to fresh Starkey-S⁰ liquid media and incubated at 30 °C. After five transfers (more than 50 generations), the cultures were diluted and plated on Starkey-Na₂S₂O₃ solid media with or without kanamycin and incubated at 30 °C. Ten days later, the percentage of plasmid maintenance was calculated by the number of colonies formed on media with kanamycin divided by that formed on media without kanamycin. The presence of the *pfkA* gene insert was analyzed by PCR with the *pfkA* oligonucleotides on isolated transconjugants.

Phosphofructokinase assays

Phosphofructokinase specific activity was assayed as described (Kotlarz & Buc 1982). Protein concentrations were determined by the method described previously (Darbre 1986) using bovine serum albumin as a standard.

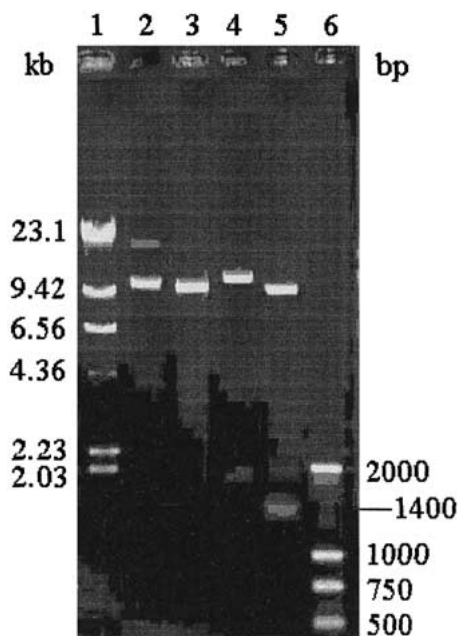


Fig. 1. Agarose gel electrophoresis (0.8% w/v) showing the plasmids pJRD215 and recombinant plasmid pSDK-1 with *pfkA* gene digested by *Bam*HI and *Hind*III. Lane 1: λ DNA/*Hind*III marker; lane 2: pJRD215 digested by *Hind*III; lane 3: pJRD215 digested by *Bam*HI and *Hind*III; lane 4: pSDK-1 digested by *Hind*III; lane 5: pSDK-1 digested by *Bam*HI and *Hind*III; lane 6: DL2000 DNA marker.

Reverse transcription PCR (RT-PCR) amplification

To verify the expression of *pfkA* gene in *A. thiooxidans*, RT-PCR was performed. Total RNA (1 μ g) was reverse-transcribed using a ThermoScript RT-PCR kit from Invitrogen. PCR amplification was carried out with primer 1 (5'-CAGCAGCAGATCGATAGCGT-3') and primer 2 (5'-CAGCAGCAGATCGATAGCGT-3'). The PCR conditions were: 5 min at 94 °C, 35 cycles of 1 min at 94 °C, 1 min at 55 °C, 1 min at 72 °C, and finally 7 min at 72 °C.

Effect of glucose on the growth of *A. thiooxidans*

A. thiooxidans was grown in Starkey-S⁰ liquid media in the absence or presence of 5 g glucose l⁻¹. The dry weight of cell suspensions was determined by filtrating culture samples over nitrocellulose filters (pore diameter 0.45 μ m). The cells were washed three times with de-ionized water, dried to constant weight at 70 °C and weighed. The concentration of glucose in culture supernatant was measured with SBA-40C Biosensor.

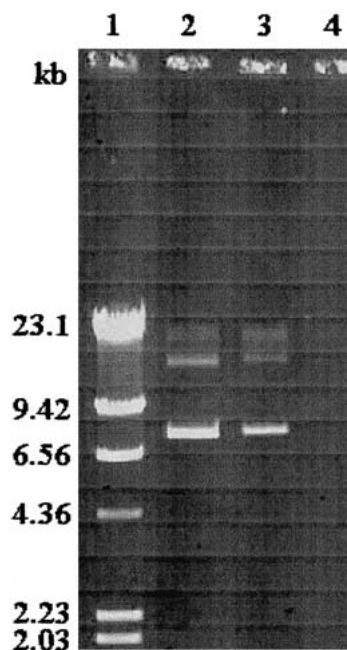


Fig. 2. Agarose gel electrophoresis (0.8% w/v) showing recombinant plasmid pSDK-1 in donor cells and transconjugant strains. Lane 1: λ DNA/*Hind*III marker; lane 2: *E. coli* SM10 donor containing recombinant plasmid pSDK-1; lane 3: *A. thiooxidans* Tt-7 transconjugant; lane 4: Tt-7 control.

Results and discussion

The agarose gel electrophoresis of plasmid pJRD215 and pSDK-1 digested by *Bam*HI and *Hind*III showed that pSDK-1 carried the 1.4 kb heterogeneous fragment (Figure 1). Sequence data confirmed that the 1.4 kb insert fragment in pSDK-1 has a sequence corresponding to that of the *pfkA* gene.

Though there are marked differences in their growth conditions, the mating medium described above could provide energy sources for either *E. coli* or *A. thiooxidans*. Using *E. coli* SM10 (pSDK-1) as the donor and *A. thiooxidans* Tt-7 as the recipient, the plasmid pSDK-1 could be mobilized into *A. thiooxidans* strain with the aid of *tra* gene integrated in the genomic DNA of *E. coli* SM10. The presence of pSDK-1 in *A. thiooxidans* transconjugants was verified by plasmid isolation (Figure 2). The transfer frequency of plasmid pSDK-1 from *E. coli* SM10 to Tt-7 strain was $1.4 \pm 0.6 \times 10^{-5}$ and the spontaneous mutation rate was lower than 10^{-8} . More than 75% of Tt-7 cells carried the pSDK-1 after being cultured for 50 generations without selective pressure, which showed that pSDK-1 was maintained consistently in *A. thiooxidans*. The PCR analysis of isolated

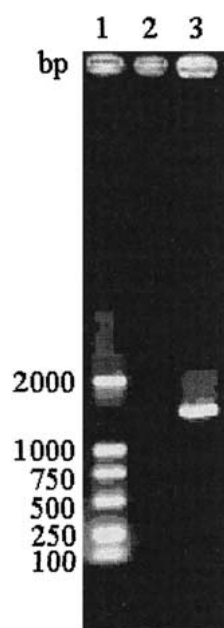


Fig. 3. PCR for detecting the presence of *pfkA* gene insert on isolated Tt-7 (pSDK-1) transconjugants after being cultured for 50 generations without selective pressure as described in materials and methods. Lane 1: DL2000 DNA marker; lane 2: PCR amplification on Tt-7, control; lane 3: PCR amplification on Tt-7 (pSDK-1) transconjugants.

transconjugants showed the presence of the *pfkA* gene in *A. thiooxidans* (Figure 3).

Table 2. Phosphofructokinase-1 activity in different strains. Specific activities were measured in the cell-free crude extract. The cells were resuspended in 50 mM Tris (pH 7.4), 1 mM dithiothreitol, 1 mM EDTA, 1 mM PMSF, broken by being frozen, thawed, treated with a ultrasonic oscillator (20 kHz, 350 W) for 20 min at 0 °C, and then centrifuged at $15\,000 \times g$ for 30 min at 4 °C to remove the cell debris. Enzyme activity units are defined as described (Kotlarz & Buc 1982).

Plasmid	Host	Specific activity ^a (U g ⁻¹)
–	<i>E. coli</i> K12	86 ± 1.9
–	DF1010	0
–	Tt-7	0
pJRD215	DF1010	0
pSDK-1	DF1010	97 ± 1.6
pJRD215	Tt-7	0
pSDK-1	Tt-7	18 ± 0.5

^aValues were obtained from three independent experiments and are shown as means ± SD.

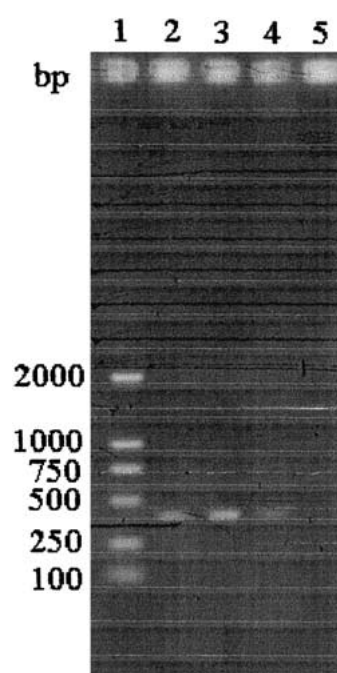


Fig. 4. RT-PCR for detecting phosphofructokinase-1 mRNA from different strains. Total RNA (1 µg) was applied to each reaction. Lane 1: DL2000 DNA marker; lane 2: *E. coli* K12; lane 3: *E. coli* DF1010 containing recombinant plasmid pSDK-1; lane 4: *A. thiooxidans* Tt-7 containing pSDK-1; lane 5: Tt-7 control.

The specific activity of phosphofructokinase-1 was detectable in *A. thiooxidans* transconjugant (Table 2). Agarose gel electrophoresis of RT-PCR products confirmed the expression of *pfkA* gene in this organism (Figure 4). These results indicated that the promoter sequences from *E. coli* could be recognized by *A. thiooxidans*. Study showed that *icd* gene from *A. thiooxidans* could function and complement a glutamate requirement of *E. coli* cells (Inoue *et al.* 2002). Some genes originated from heterotrophic bacteria, including the kanamycin, streptomycin, tetracycline resistance genes, *tra* gene of RP4 and arsenic resistance gene, have been cloned and expressed in *A. thiooxidans* (Jin *et al.* 1992, 1993, Shao *et al.* 1997). The gene expression system therefore might be similar in *A. thiooxidans* and *E. coli* though there are great physiological differences between them.

The enzyme activity in Tt-7 (pSDK-1) was unexpectedly low possibly because the promoter cloned from *E. coli* was not recognized efficiently in *A. thiooxidans*. Previous studies have shown that the *rbcLS* genes as well as the *recA* and *alaS* genes from another members of the *Acidithiobacillus* genus *A. ferrooxidans* were not expressed in *E. coli* unless placed

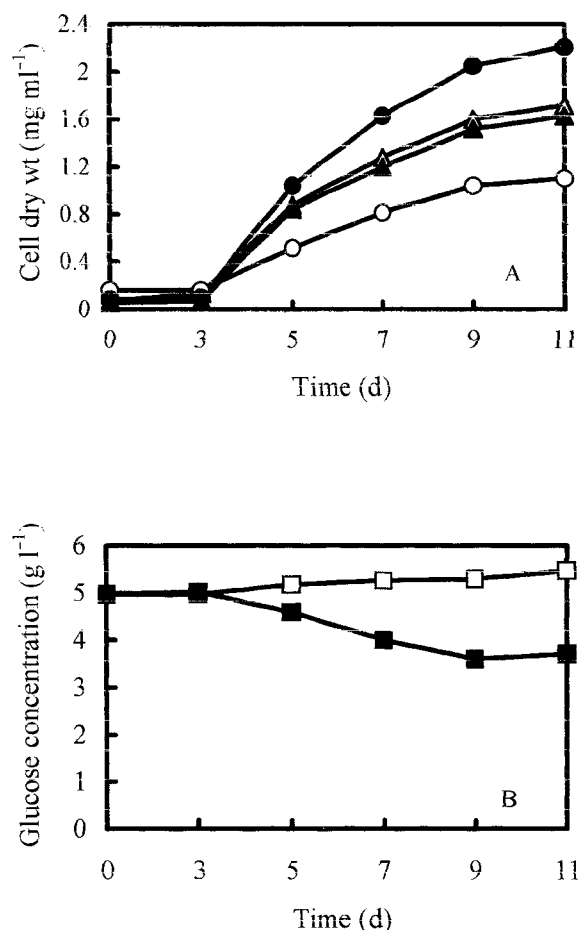


Fig. 5. (A) Growth curves of *A. thiooxidans* Tt-7 and its transconjugant Tt-7 (pSDK-1) strain in Starkey-S⁰ inorganic medium with or without 5 g glucose l⁻¹ at 30 °C. Δ, Tt-7, growth in the absence of glucose, control; ▲, Tt-7, growth in the presence of glucose; ○, Tt-7 (pSDK-1), growth in the absence of glucose; ●, Tt-7 (pSDK-1), growth in the presence of glucose. (B) Comparison of glucose concentration in cultures of Tt-7 (□) and Tt-7 (pSDK-1) (■) during the whole cultivation. Each data point represents the mean value of three independent experiments.

under the control of an *E. coli* promoter (Kusano *et al.* 1991, Guiliani *et al.* 1997). In another research, the *A. ferrooxidans ntrA* gene product was able to promote σ^{54} -dependent transcription in *E. coli ntrA* mutants. But complementation was only partial and insufficient to permit growth of *E. coli ntrA* mutants (Berger *et al.* 1990). Therefore some differences might exist in the gene expression systems between *Acidithiobacillus* and *E. coli*.

Of great significance was that glucose significantly stimulated the cell growth of the *A. thiooxidans* Tt-7 transconjugants but had no effect on the Tt-7 (Figure 5A). Figure 5B shows that the concentration of

glucose in the growth medium did not change for Tt-7 but decreased gradually for Tt-7 (pSDK-1) during the whole cultivation, however, the concentration of glucose decreased very slowly. The growth of autotrophic *A. thiooxidans* was not influenced by glucose whereas the growth of transconjugants was stimulated by the consumption of glucose though their ability to consume the glucose was limited. These results suggest that lack of the key enzyme activity of glucose metabolic pathway in Tt-7 causes its inability to consume glucose. Whereas the expression of *pfkA* gene in the transconjugant caused assimilation of glucose to the synthesized cell carbon, since the fixation of CO₂ has a high energy requirement, synthesis of a part of the cell material from glucose instead of CO₂ should have an energy sparing effect which should lead to an increase in cell yield. But the lack of a complete Krebs cycle and low activities of key enzymes of the cycle might prevent the metabolism of glucose at a fast rate enough to satisfy major energetic and biosynthetic needs.

Since the industrial utilization of *A. thiooxidans* has been limited by the slow growth rate and the low cell yield of this organism, the faster growth strains constructed would be useful in the metal leaching from mineral ores and in the microbial desulfurization of coal. Our work may provide a possible way to develop new strains with better growth rates and increased leaching efficiencies.

Acknowledgements

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